Solution Requirements

Date	02 October 2026
Team ID	SWTID-2026-3981
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users should have a lightweight session-based setup or basic login system for application access.	Medium
2	Authorization levels	Student/Learner role should have access to normal Q&A, quiz, and roadmap features; Admin role should have permission to monitor system status.	Low

3	External interfaces	Google Gemini 1.5 Pro API (google-generativeai SDK) ki should connect to process advanced AI queries, summaries, and quizzes.	High

4	Transactions processing	User input prompts should be processed quickly; JSON responses from FastAPI endpoints (/qa, /explain, /quiz, /summarize, /learn/recommendations) should be validated and rendered.	High
5	Reporting	Active learning path recommendation summaries, quiz scores, and generated summaries should be displayed on the dashboard in a presentation-ready format.	High
6	Business rules, etc.		High
7	Compliance to laws or regulations	Academic integrity constraints should be followed; safe content generation filters should be applied and user input data privacy should be maintained.	Medium
8	Other	Multi-functional REST API architecture should support reusability for future mobile/web clients.	High

Step 2: Non-Functional Requirements (NFR)

S. No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	API response latency should be minimal and the UI layout should render quickly.	API response time < 3 seconds for cloud queries; page load < 1.5 seconds.
2	Scalability	Asynchronous FastAPI architecture should efficiently process multi-request workloads.	Handles up to 100+ concurrent user requests without server crash.
3	Security & Data Privacy	External Gemini API keys should be safely isolated in environment variables.	100% secure API key storage using environment variables.
4	Reliability & Availability	With offline fallback model integration, basic AI responses should remain functional during network issues.	99.5% uptime with dual-model failover support.
5	Usability & Accessibility	Simple, clean HTML/CSS web interface should be responsive across standard desktop and mobile screen dimensions.	Responsive UI rendering across standard modern desktop & mobile browsers.
6	Other	Application should run seamlessly on standard CPU hardware and Mac M1/Apple Silicon without requiring high-end servers.	Runs on Standard computing environments with 8GB RAM without

			GPU dependence.